

GODERICH, ON, Canada

WMO: 712610

Lat: 43.7647N

Lon: 81.7122W

Elev: 214

StdP: 98.78

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-14.8	-20.6	0.6	-16.8	-18.0	0.8	-14.2	13.4	-5.2	12.4	-6.2	4.4	80	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	28.5	22.2	27.0	21.4	25.7	20.7	23.7	26.5	22.7	25.4	21.8	24.5	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.0	25.4	21.8	16.8	24.3	20.8	15.9	23.5	72.0	26.5	68.0	25.5	64.7	24.6	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.5	9.4	-21.7	30.8	3.4	1.2	-24.1	31.7	-26.1	32.5	-28.0	33.2	-30.5	34.1		
			-21.9	25.3	3.4	1.4	-24.3	26.3	-26.3	27.1	-28.2	27.9	-30.7	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-4.5	-4.3	-0.3	5.9	12.3	17.5	19.7	19.2	16.1	10.4	4.3	-1.0
	DBStd	9.89	5.76	5.64	5.86	4.94	4.84	4.10	3.17	3.06	3.90	4.46	4.49	4.54
	HDD10.0	1919	451	401	326	142	30	1	0	0	2	49	179	340
	HDD18.3	3963	709	634	579	372	197	65	23	27	88	248	422	598
	CDD10.0	1192	1	0	4	20	102	225	301	284	186	62	7	1
	CDD18.3	193	0	0	0	1	10	39	64	53	23	3	0	0
	CDH23.3	1022	0	0	1	3	64	236	341	249	117	11	0	0
	CDH26.7	155	0	0	0	0	8	43	51	38	14	1	0	0
Wind	WSAvg	4.7	5.8	5.3	4.9	4.9	4.2	3.7	3.4	3.5	3.9	4.9	5.6	5.8
Precipitation	PrecAvg	992	94	71	66	72	85	82	86	74	91	86	94	91
	PrecMax	1275	154	124	150	136	176	177	155	161	238	167	167	181
	PrecMin	701	60	33	27	24	32	41	23	23	21	23	40	53
	PrecStd	149	25	26	31	30	33	35	34	28	45	35	31	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	11.9	19.2	23.1	27.7	30.1	30.0	29.9	28.5	24.9	17.8	13.1
		MCWB	10.8	10.2	14.7	17.0	19.9	22.8	23.8	23.3	21.8	19.0	13.6	11.9
	2%	DB	7.8	7.7	15.0	19.6	25.1	27.8	28.1	27.5	26.3	22.0	15.0	9.3
		MCWB	6.7	6.1	11.3	14.0	18.6	21.7	22.5	22.1	20.8	17.3	12.4	8.1
	5%	DB	4.7	4.8	10.8	16.9	22.9	26.1	26.5	25.9	24.6	19.7	13.0	7.0
		MCWB	3.7	3.2	8.1	11.9	17.0	20.4	21.5	21.3	19.8	16.0	10.9	5.9
	10%	DB	2.7	2.6	7.6	14.2	20.3	24.2	25.2	24.5	22.8	17.4	10.9	4.8
		MCWB	1.8	1.4	5.4	10.1	15.3	19.5	20.8	20.7	19.0	14.5	9.0	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.9	15.2	18.4	21.5	24.3	25.6	25.0	23.5	20.4	15.0	11.7
		MCDB	11.9	11.8	18.5	21.7	26.5	28.2	28.6	28.0	26.8	23.5	16.6	12.8
	2%	WB	6.9	6.2	11.9	15.0	19.5	22.7	23.7	23.4	21.9	18.3	12.9	8.5
		MCDB	7.5	7.3	14.5	18.2	23.6	26.4	26.4	25.9	24.6	20.9	14.4	9.2
	5%	WB	3.9	3.4	8.6	12.7	18.0	21.5	22.6	22.3	20.7	16.3	11.0	5.9
		MCDB	4.6	4.6	10.6	16.1	21.5	25.0	25.1	24.7	23.3	19.0	12.6	7.1
	10%	WB	1.8	1.5	5.6	10.4	16.1	20.3	21.7	21.4	19.5	14.7	9.1	3.6
		MCDB	2.5	2.5	7.5	13.8	19.5	23.2	24.2	23.7	22.2	17.2	10.9	4.7
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.7	7.3	8.8	9.2	8.7	8.5	8.5	9.0	7.5	5.9	5.0
		MCDBR	7.7	8.4	11.1	13.0	11.6	10.5	10.0	9.7	10.5	10.2	8.8	7.3
		MCWBR	7.3	7.6	8.5	8.8	6.7	6.0	5.8	5.8	6.2	6.6	7.4	6.9
	5% WB	MCDBR	7.6	8.3	10.5	11.9	10.2	9.2	8.2	8.0	8.7	9.1	8.0	7.3
		MCWBR	7.5	7.8	8.6	9.1	7.1	5.9	5.5	5.6	6.0	6.5	7.3	7.1
Clear-Sky Solar Irradiance	taub	0.294	0.319	0.327	0.361	0.387	0.422	0.409	0.402	0.376	0.358	0.313	0.293	
	taud	2.410	2.274	2.360	2.333	2.326	2.253	2.323	2.324	2.392	2.411	2.508	2.461	
	Ebn at Noon	836	866	912	902	884	851	859	852	849	815	811	805	
	Edh at Noon	75	102	107	120	124	134	124	120	104	89	69	66	
All-Sky Solar Radiation	RadAvg	1.23	2.12	3.47	4.54	5.45	6.06	6.01	5.21	4.12	2.46	1.41	0.91	
	RadStd	0.18	0.26	0.41	0.41	0.47	0.35	0.33	0.26	0.37	0.24	0.22	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	-0.84	-1.10	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page